

Graduation Power BI Project – Step-by-Step Documentation

1. Project Overview

- **Project Name:** Bank Loan
 - **Tool Used:** Power BI Desktop
 - **File Type:** .pbix
 - **Objective:** Build an analytical dashboard including revenue, orders, and rating metrics.
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2. Data Acquisition

Step 2.1 – Import PBIX File

- Opened the provided .pbix file in Power BI Desktop.
- Ensured correct loading without errors.

Step 2.2 – Identify Connected Data Sources

- Reviewed data source settings from:
File → Options & settings → Data source settings.

Step 2.3 – Validate Access Permissions

- Checked that all data sources were accessible and authenticated properly.
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3. Data Transformation (Power Query Editor)

Step 3.1 – Open Power Query

- Navigated to:
Home → Transform Data.

Step 3.2 – Review Existing Queries

- Inspected all tables and applied transformation steps in the query pane.

Step 3.3 – Standard Cleanup Operations

Typical operations documented:

Action	Purpose
Remove duplicates	Ensure record uniqueness
Fill missing values	Maintain data consistency
Change data types	Enable proper calculations and relationships
Remove errors / null rows	Clean corrupted or incomplete records
Rename columns	Improve clarity and semantic meaning
Create calculated columns (if any) Add derived values before load	

Step 3.4 – Close & Apply

- Clicked:
Home → Close & Apply to load clean data into the model.

4. Data Modeling

Step 4.1 – Review Tables

- Went to:
Model View.
- Reviewed schema and verified table structure.

Step 4.2 – Create/Validate Relationships

Documented for every relationship created or modified:

From Table	Column	To Table	Column	Type	Direction
Customers	CustomerID	Orders	CustomerID	One-to-Many	Single

Orders	OrderID	Reviews	OrderID	One-to-Many	Single
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Step 4.3 – Apply Star/Snowflake Schema Rules

- Ensured fact tables contain numeric fields.
 - Ensured dimension tables contain descriptive attributes.
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5. DAX Measures Created

Each measure documented individually:

Measure Name	DAX Formula	Description
Total Revenue	SUM(Orders[Revenue])	Sum of all sales
Total Orders	COUNT(Orders[OrderID])	Count of orders
AOV (Avg Order Value)	DIVIDE([Total Revenue], [Total Orders])	Average revenue per order
Net Revenue	CALCULATE([Total Revenue], Orders[Status] = "Delivered")	Revenue only from completed deliveries
Review Rate	AVERAGE(Reviews[Rating])	General average rating

6. Report on Development

Step 6.1 – Create Pages

Documented pages added:

Page Name	Purpose
Revenue Overview	KPI summary for revenue and AOV
Customer Ratings	Insights per customer/product
Order Details	Order completion and counts

Step 6.2 – Add Visuals

Examples documented:

Visual Type	Fields Used	Purpose
Card	Total Revenue	Show KPI
KPI Visual	Net Revenue	Show delivered revenue
Table	Orders with status	View details
Bar Chart	Products vs Rating	Compare product reviews
Line Chart	Revenue over time	Trend analysis

Step 6.3 – Apply Filters

Documented filters:

Filter Level	Field	Type
Page	Date	Between
Page	Order Status	Equals
Report	Customer	Dropdown

7. Testing & Validation

Step 7.1 – Validate Measures

Checked:

- Totals match expected results.
- No division errors using DIVIDE().

Step 7.2 – Validate Relationships

Confirmed:

- No circular relationships.
- Correct cardinality.

Step 7.3 – Refresh Test

- Used:
Home → Refresh.

8. Deployment Preparation

Step 8.1 – Optimize PBIX Size

- Removed unused columns from Power Query.
- Ensured only required tables loaded.

Step 8.2 – Add Documentation in Power BI

- Added dataset description from:
Model View → Properties → Description.

9. Final Output

- Dashboard ready for sharing via Power BI Service or presentation.
- .pbix saved without errors.